

Towards Situational Awareness of Urban Pedestrian Flows: An Assessment of Data Quality on the Performance of Predictive Models

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Summary

This study assesses IoT sensor data's impact on urban pedestrian flow prediction. It contrasts univariate and multivariate LSTM models, revealing the latter's superior performance in accuracy and anomaly detection, especially with incomplete data. Findings advocate for matching data quality with predictive techniques to optimise urban pedestrian flow management.

KEYWORDS: Geospatial, Anomaly Detection, Pedestrians, IoT, Machine Learning

1. Introduction

This paper investigates the influence of data quality from Internet of Things (IoT) sensors on forecasting pedestrian movements in urban settings. As cities grow in complexity, understanding people mobility and flows becomes pivotal to city management (Batty, 2007). This study seeks to understand both the benefits and potential constraints of utilising IoT sensor data to enhance our foresight and handling of urban flows such as pedestrian movements. Central to this research is situational awareness—the ability to perceive and predict environmental dynamics. This capability is crucial in managing urban intricacies where timely, informed decisions can greatly impact outcomes, for example emergency response scenarios (Xu *et al.*, 2023). The broader goal is to contribute to the growing research into real-time pedestrian simulation (Ternes *et al.*, 2022; Malleon *et al.*, 2022), a critical part of the wider urban digital twin programme (CDBB, 2020). To this end artificial intelligence (AI) and big data analytics are employed to refine decision-making in real-time in relation to the challenge of short-term crowd movement predictions.

2. Research Objectives

A cloud-based system is presented that rapidly identifies atypical data patterns, thereby facilitating swift and effective responses to events, such as a potential overcrowding. Key research questions investigated are:

- 1) How does IoT sensor data quality impact predictive accuracy?
- 2) Can irregularities in crowd movements be predicted using unsupervised methods, and what do these patterns tell us?
- 3) Do additional data types such as periodicity bolster the model's predictive power?
- 4) How does variable data quality affect model outcomes?

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3. Methodology

To address the research questions, a methodology involving two distinct models was developed: a Long Short-Term Memory (LSTM) network and a linear regression model. The LSTM was selected for its proficiency in capturing time-dependent patterns, while the linear model provided a benchmark for comparison due to its simplicity and interpretability (Jing *et al.*, 2021; Siami-Namini, Tavakoli and Siami Namin, 2018; Fu, Zhang and Li, 2016).

The data employed in study originated from IoT sensors located on Northumberland Street (**Figure 1**), Newcastle upon Tyne collected by the Newcastle University [Urban Observatory](#) – an extensive centralised urban data repository. This primary source was supplemented by university term dates, providing a contextual backdrop for pedestrian flow patterns. The dataset encompasses three years of pedestrian activity, recorded at 15-minute intervals, and is approximately 85% complete (**Figure 2**). The data has known uncertainties inherited from the data extraction algorithms that have been noted but not considered in this research. **Table 1** shows the edge-processed data sent from the sensor to the database. Each value denotes the number of people that passed the sensor during a 15-minute period. The two sensors explored in this research (east and west) are co-located, facing in opposite directions on the same street. Each model is only trained on either the east or west dataset. The full methodology describing data-preprocessing and limitations of this study can be found [here](#).

Table 1 Sample data from an Urban Observatory pedestrian sensor on Northumberland Street, Newcastle upon Tyne.

dt	value	veh_class	dir	location	category
01/12/2022 00:00	13	person	northwest_to_southwest	NclNorthumberlandStSavilleRowWest	flow
01/12/2022 00:00	15	person	southwest_to_northwest	NclNorthumberlandStSavilleRowWest	flow
01/12/2022 00:15	20	person	northwest_to_southwest	NclNorthumberlandStSavilleRowWest	flow
01/12/2022 00:15	8	person	southwest_to_northwest	NclNorthumberlandStSavilleRowWest	flow



Figure 1 A pedestrian sensor performing object detection.

Figure 2 shows the raw data over the course of the sixteen months investigated. At 0%, every record made by the sensor is represented. Each day, each sensor should be recording 96 observations – $4 * 15\text{-minutes} * 24 * \text{hours}$. **Figure 2** shows how removing data using different thresholds of daily completeness affects the structure of the data. At 100% (96 records per day) and 94% (min 90 records per day) there is a notable difference in data sparsity - very few days have the full 96 records, but many have at least 90 records.

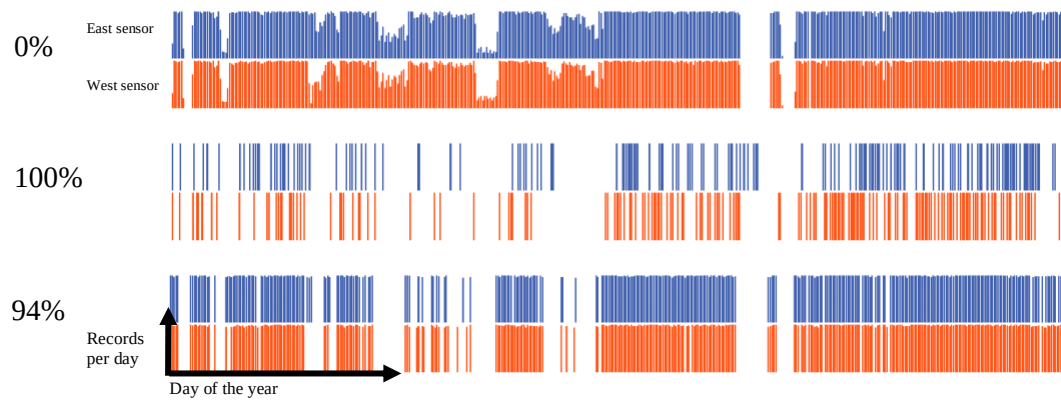


Figure 2 Data completeness over 16 months.

Univariate Modelling: The univariate modelling used sequences (varying in length from between one hour and 48 hours) of pedestrian flow data to predict subsequent flow. This model was tasked to forecast between one and 6 hours into the future using the past 48 data points. This simplistic model served as a foundation for understanding the potential and limitations of using raw data streams for prediction.

Multivariate Modelling: In contrast, the multivariate model integrated additional variables, such as sine functions representing time, to offer a more complex and potentially insightful analysis. The model's input consisted of multiple features, including engineered ones that encapsulated periodicity and external factors affecting pedestrian movement. Some of the engineered features were extracted using the Lomb-Scargle method to extract key signal frequencies in the data. **Figure 3** shows that the dominant frequency is the daily period, followed by the diurnal, annual and quarterly periods. The frequencies are encoded as waves and presented to the model as additional features to learn.

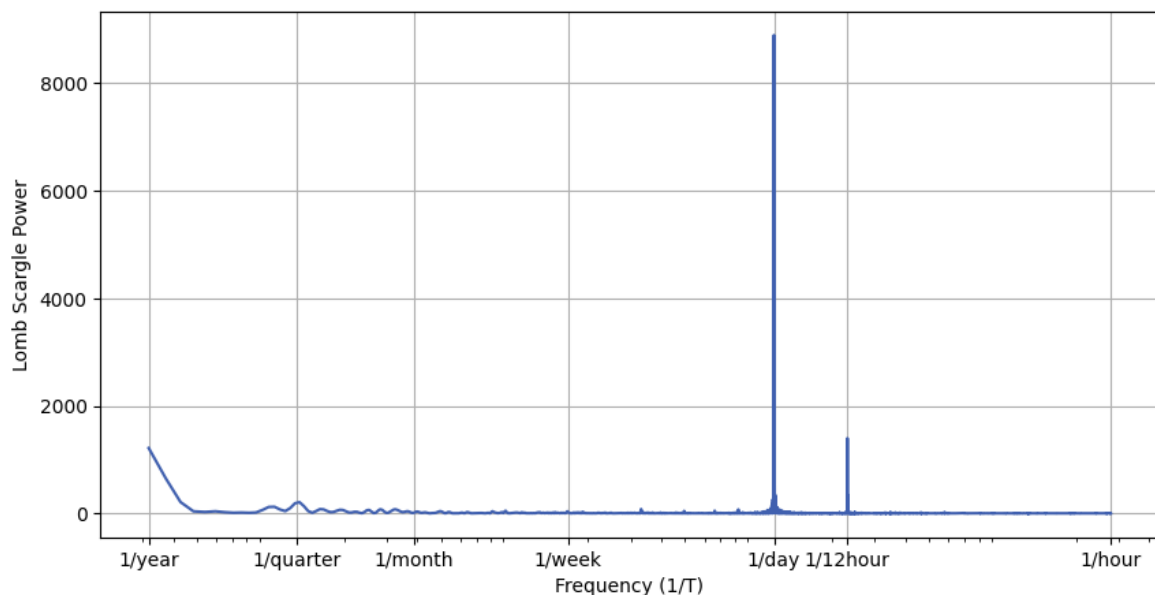


Figure 3 Lomb-Scargle Periodogram

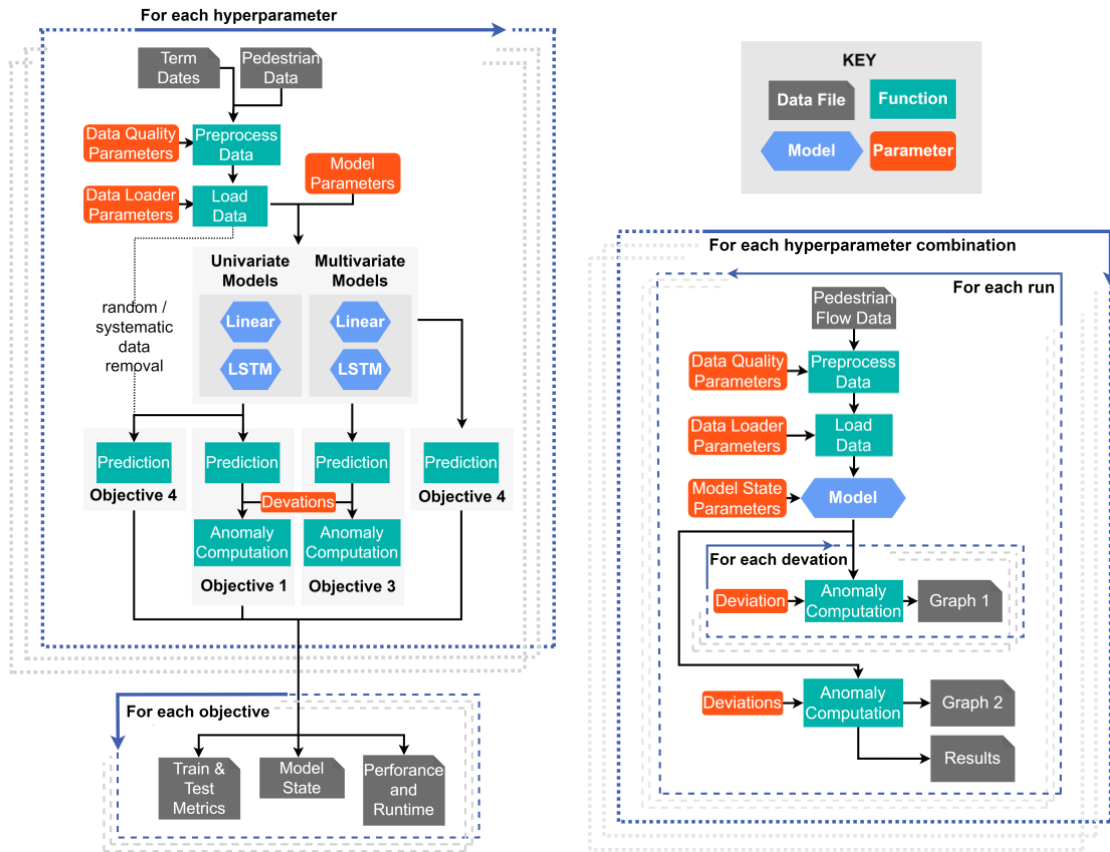


Figure 4 Modelling workflow – training left, evaluation right.

Modelling Workflow: Both models underwent a training process, iterating over numerous epochs to optimise their predictive capabilities. The linear model revealed the relative importance of each input variable when tasked with predicting pedestrian counts. **Figure 4** shows the full modelling workflow for each of the research objectives.

Anomaly Detection: Central to the analysis was the detection of anomalies. Single-step prediction strategies were employed to ascertain whether the models could identify unusual patterns in unseen data. Anomalies were defined as deviations from the established patterns that could signal potential issues within the urban flow, such as overcrowding or unusual dispersal of pedestrians.

Predictive Horizon and Data Robustness: The impact of extending the predictive horizon on accuracy was explored with the expectation that longer horizons would yield greater error rates. Additionally, model robustness was assessed by varying data completeness levels, to emulate real-world scenarios where sensor data might be intermittently missing.

4. Results

The univariate models struggled with the complexity of predicting pedestrian flow. While they could forecast movements over a period of a few hours, their accuracy deteriorated as the prediction horizon extended beyond 12 hours. This was particularly evident when these models encountered anomalous data points which influenced subsequent predictions, thereby highlighting the models' limited adaptability to sudden changes in data patterns. **Figure 5** shows the rapid increase in MAE (mean absolute error) as the prediction horizon increases up to 6 hours in the future and for training windows of different lengths.

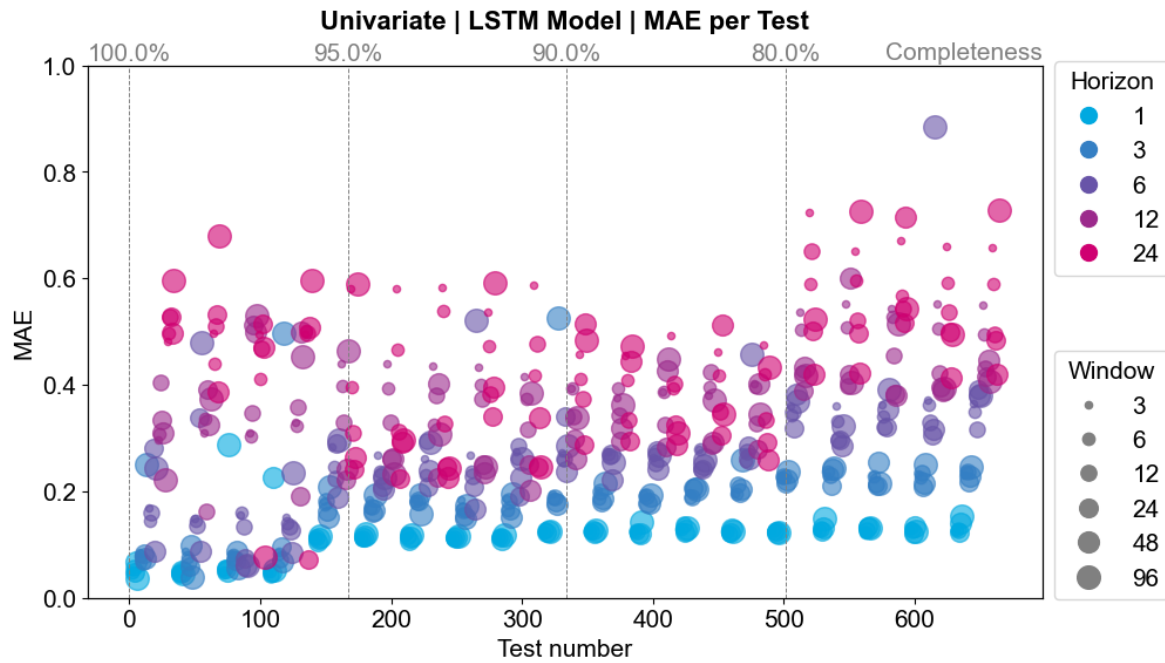


Figure 5 Univariate LSTM prediction results.

In contrast, the multivariate models demonstrated a significant improvement in prediction accuracy, with MAE dropping from ≈ 0.8 in the univariate to ≈ 0.4 for some of the models with the longest prediction horizons. The addition of extra variables allowed these models to discern more complex patterns and relationships within the data. The engineered features, representing periodicity, provided the models with essential temporal context that improved their forecasting ability, particularly in data sets with lower completeness thresholds. **Figure 6** shows the improvement compared to the univariate model of **Figure 5**; MAE is much lower at the longest horizon of 6 hours and the multivariate model performs well on less complete data.

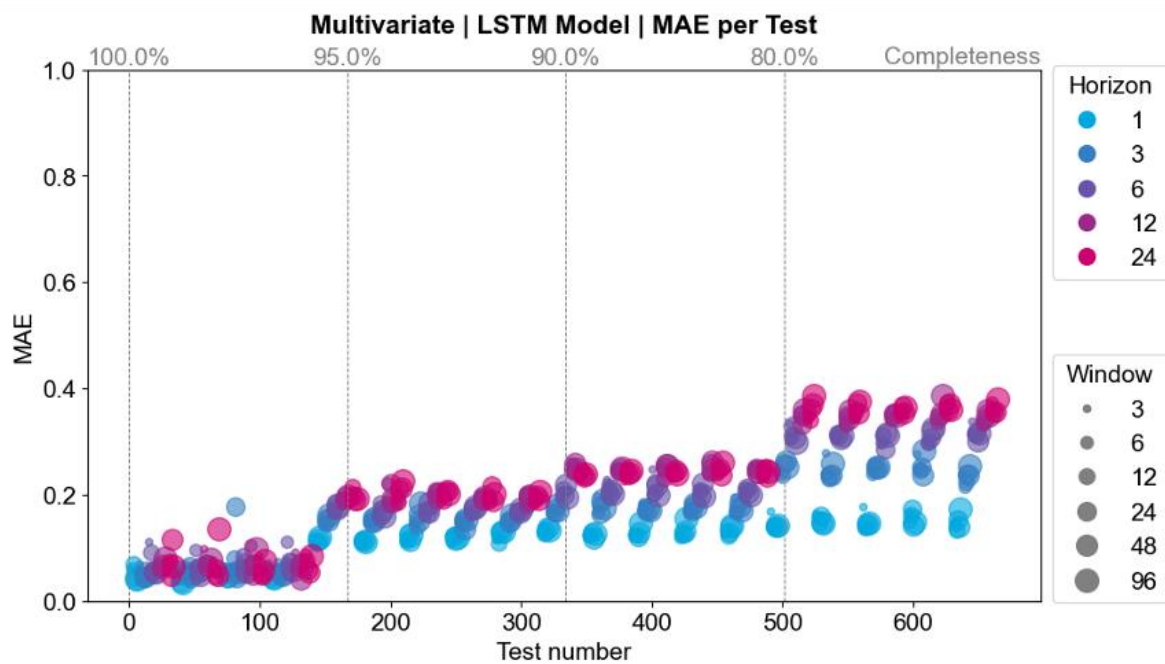


Figure 6 Multivariate LSTM model performance.

Both models showed that a longer prediction horizon, and a lower completeness of data increases error rates. However, the multivariate models showcased a degree of resilience, maintaining a relatively stable error rate across varying horizons, thus affirming their suitability for practical applications like real-time crowd management. Robustness tests revealed that models trained on datasets with a 100% completeness threshold yielded the highest accuracy for short prediction horizons.

In the case of anomaly detection, the multivariate models outperformed their univariate counterparts, albeit not as significantly as in the prediction accuracy seen above. This finding suggests that while extra data types improve forecasting, they do not necessarily translate into a commensurate improvement in anomaly detection (a full explanation can be found [here](#)).

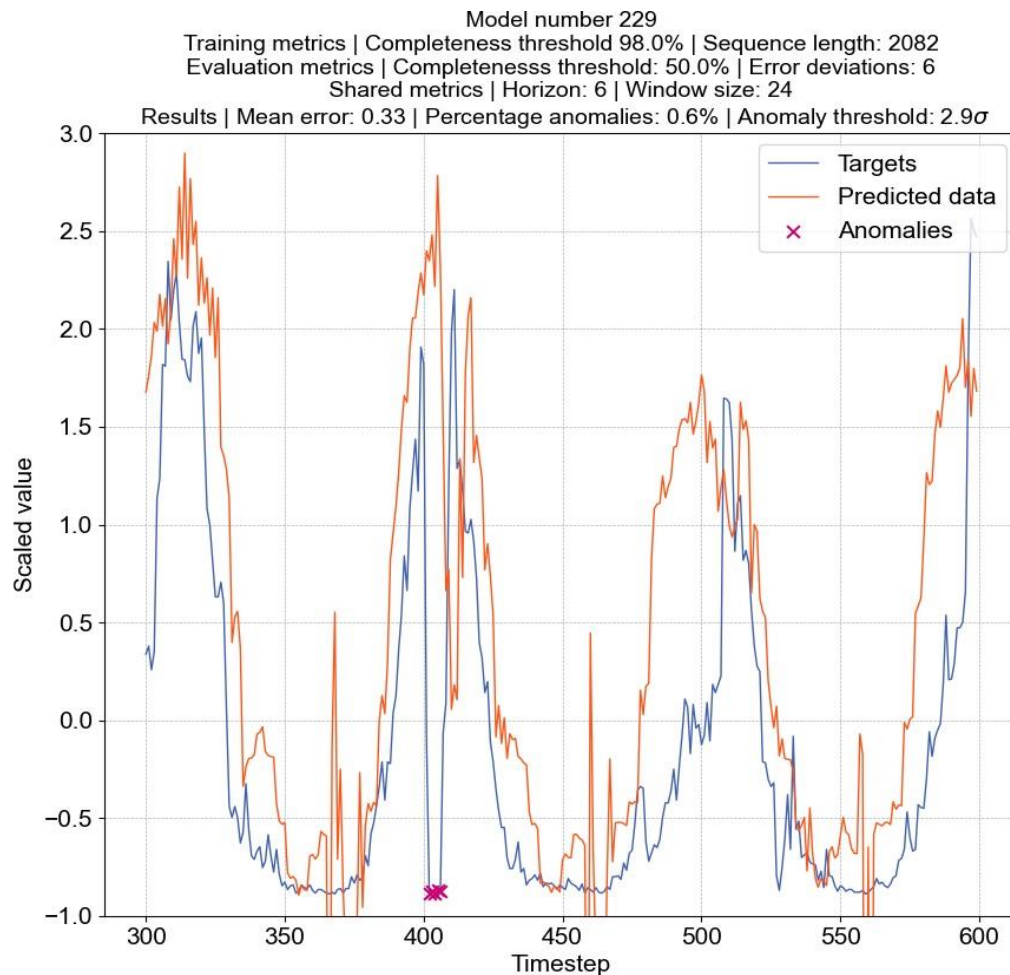


Figure 7 Anomaly detection example model.

Figure 7 shows an example of anomalies being correctly identified. These are sequential anomalies, that unlike outliers, fall within the normal range, but their occurrence is an unusual sequence where there are very few people at midday, usually the busiest period. It is not clear whether these data points are anomalous or erroneous (for example a large object temporarily blocking the sensor), but flagging such values is a critical first step in identifying both cases.

5. Summary

The research demonstrated that univariate LSTM models are limited in their capacity to handle complex urban pedestrian dynamics, particularly when predicting over extended horizons. The multivariate LSTM model, with additional data types and engineered features, significantly reduce prediction error and enhance robustness with regards to incomplete data. However, the relationship between prediction horizon and accuracy is not straightforward; multivariate models exhibit more consistent performance across different horizons, but models needed to be trained on less complete datasets in order to have the ability to adapt to and predict with missing data, which is critical for real-world application.

Overall, multivariate models with additional data types significantly enhance predictive accuracy and robustness, particularly in the context of incomplete datasets. However, anomaly detection remains a nuanced challenge, and an area for further research. While the transformative potential of AI and big data analytics is well recognised in urban management this research highlights the importance of matching appropriate data quality with predictive modelling techniques.

References

- Batty, M. (2007) *Cities and complexity: understanding cities with cellular automata, agent-based models, and fractals*. The MIT press.
- CDBB (2020) *The approach to delivering a National Digital Twin for the United Kingdom*: Centre for Digital Built Britain.
- Fu, R., Zhang, Z. and Li, L. 'Using LSTM and GRU neural network methods for traffic flow prediction'. *2016 31st Youth Academic Annual Conference of Chinese Association of Automation (YAC)*: IEEE, 324-328.
- Jing, Y., Hu, H., Guo, S., Wang, X. and Chen, F. (2021) 'Short-Term Prediction of Urban Rail Transit Passenger Flow in External Passenger Transport Hub Based on LSTM-LGB-DRS', *IEEE Transactions on Intelligent Transportation Systems*, 22(7), pp. 4611-4621.
- Malleson, N., Birkin, M., Birks, D., Ge, J., Heppenstall, A., Manley, E., McCulloch, J. and Ternes, P. (2022) 'Agent-based modelling for Urban Analytics: State of the art and challenges', *AI Communications*, (Preprint), pp. 1-14.
- Siami-Namini, S., Tavakoli, N. and Siami Namin, A. 'A Comparison of ARIMA and LSTM in Forecasting Time Series'. 2018: IEEE.
- Ternes, P., Ward, J. A., Heppenstall, A., Kumar, V., Kieu, L.-M. and Malleson, N. (2022) 'Data assimilation and agent-based modelling: towards the incorporation of categorical agent parameters', *Open Research Europe*, 1(131).
- Xu, H., Berres, A., Yoginath, S. B., Sorensen, H., Nugent, P. J., Severino, J., Tennille, S. A., Moore, A., Jones, W. and Sanyal, J. (2023) 'Smart Mobility in the Cloud: Enabling Real-Time Situational Awareness and Cyber-Physical Control Through a Digital Twin for Traffic', *IEEE Transactions on Intelligent Transportation Systems*, 24(3), pp. 3145-3156.

Biographies

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